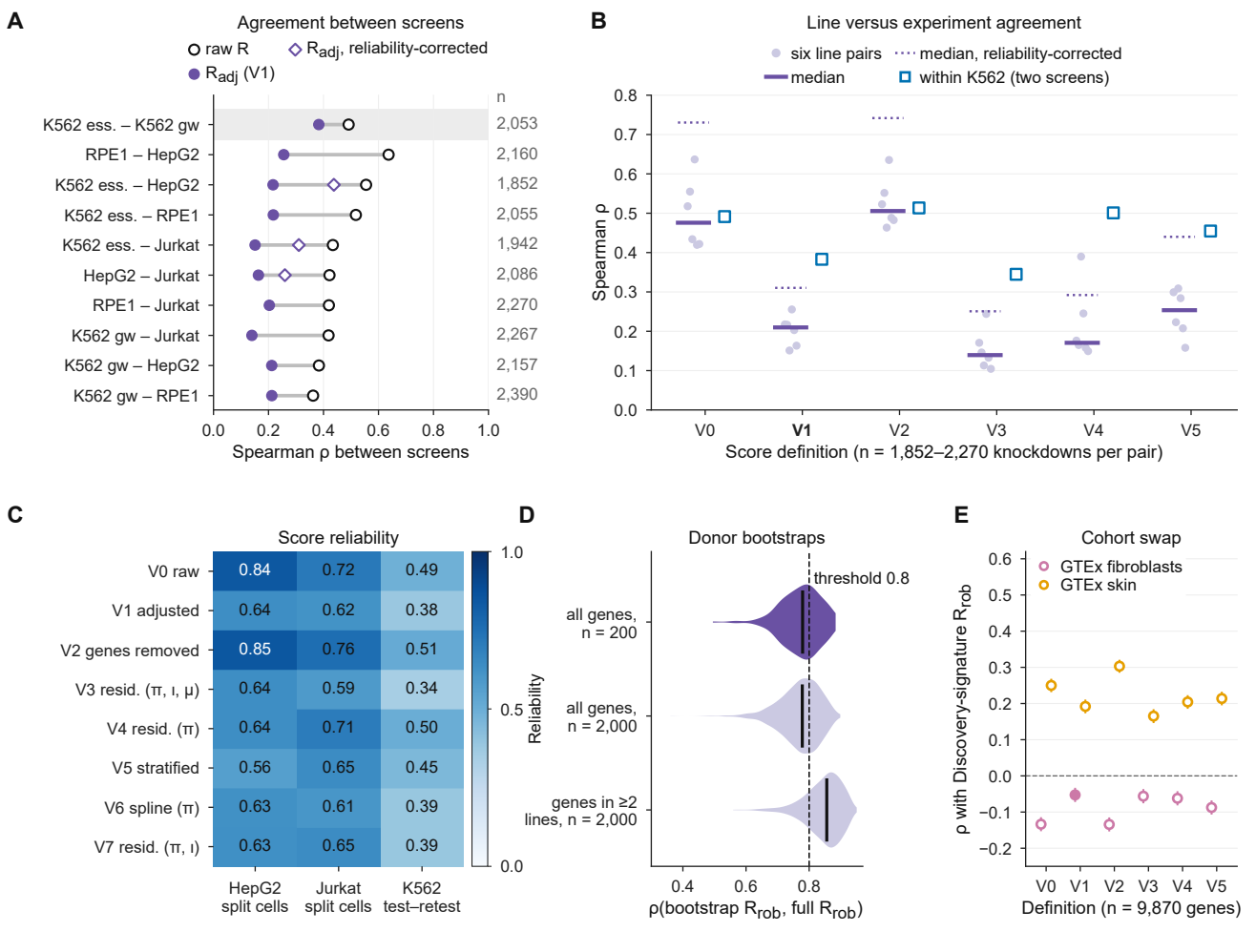

A
Agreement between screens
raw R
R_adj, reliability-corrected
R_adj (V1)
n
K562 ess. – K562 gw
2,053
RPE1 – HepG2
2,160
K562 ess. – HepG2
1,852
K562 ess. – RPE1
2,055
K562 ess. – Jurkat
1,942
HepG2 – Jurkat
2,086
RPE1 – Jurkat
2,270
K562 gw – Jurkat
2,267
K562 gw – HepG2
2,157
K562 gw – RPE1
2,390
0.0
0.2
0.4
0.6
0.8
1.0
Spearman ρ between screens
B
Line versus experiment agreement
six line pairs
median, reliability-corrected
median
within K562 (two screens)
Spearman ρ
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
V0
V1
V2
V3
V4
V5
Score definition (n = 1,852–2,270 knockdowns per pair)
C
Score reliability
V0 raw
0.84
0.72
0.49
V1 adjusted
0.64
0.62
0.38
V2 genes removed
0.85
0.76
0.51
V3 resid. (π, ι, μ)
0.64
0.59
0.34
V4 resid. (π)
0.64
0.71
0.50
V5 stratified
0.56
0.65
0.45
V6 spline (π)
0.63
0.61
0.39
V7 resid. (π, ι)
0.63
0.65
0.39
HepG2 split cells
Jurkat split cells
K562 test–retest
Reliability
1.0
0.5
0.0
D
Donor bootstraps
threshold 0.8
all genes, n = 200
all genes, n = 2,000
genes in ≥2 lines, n = 2,000
0.4
0.6
0.8
ρ(bootstrap R_rob, full R_rob)
E
Cohort swap
GTEx fibroblasts
GTEx skin
ρ with Discovery-signature R_rob
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
V0
V1
V2
V3
V4
V5
Definition (n = 9,870 genes)